

Multiple Choice (10 marks)

1. The _____ can cost you money, by sending text messages from your mobile phone numbers.
 - (a) IM - Trojans
 - (b) Backdoor Trojans
 - (c) SMS Trojan
 - (d) Ransom Trojan
2. The sub key length at each round of DES is_____
 - (a) 32
 - (b) 56
 - (c) 48
 - (d) 64
3. The four Primary Security Principles related to messages are
 - (a) Confidentiality, Integrity, Non repudiation and Authentication
 - (b) Confidentiality, Access Control, Integrity, Non repudiation
 - (c) Authentication, Authorization, Availability, Integrity
 - (d) Availability, Authorization, Confidentiality, Integrity
4. Which Nmap scan is does not completely open a TCP connection?
 - (a) SYN stealth scan
 - (b) TCP connect
 - (c) XMAS tree scan
 - (d) ACK scan
5. Man in the middle attack can endanger the security of Diffie Hellman method if two parties are not
 - (a) Joined
 - (b) Authenticated
 - (c) Submitted
 - (d) Shared

True / False (10 marks)

Answer True or False

6. True or False: In message confidentiality, both the sender and the receiver must be able to interpret the transmitted message.
7. True or False: The Deep Web consists of content not indexed by traditional search engines, such as databases and private networks.
8. True or False: The output of a hash function is typically known as a Message Authentication Control (MAC).
9. True or False: A ping sweep is primarily utilized to locate firewalls by detecting their response to network probes.
10. True or False: The IP Network Browser is a tool commonly used for performing SNMP enumeration in network security assessments.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the significance of the Tor browser in facilitating anonymous access to the Tor network, including its impact on privacy, security, and potential misuse in cybersecurity contexts.
12. Discuss the role of address sanitizer (ASAN) in enhancing the effectiveness of fuzzing techniques for identifying memory errors in software. How does ASAN facilitate the debugging process?

End of Paper